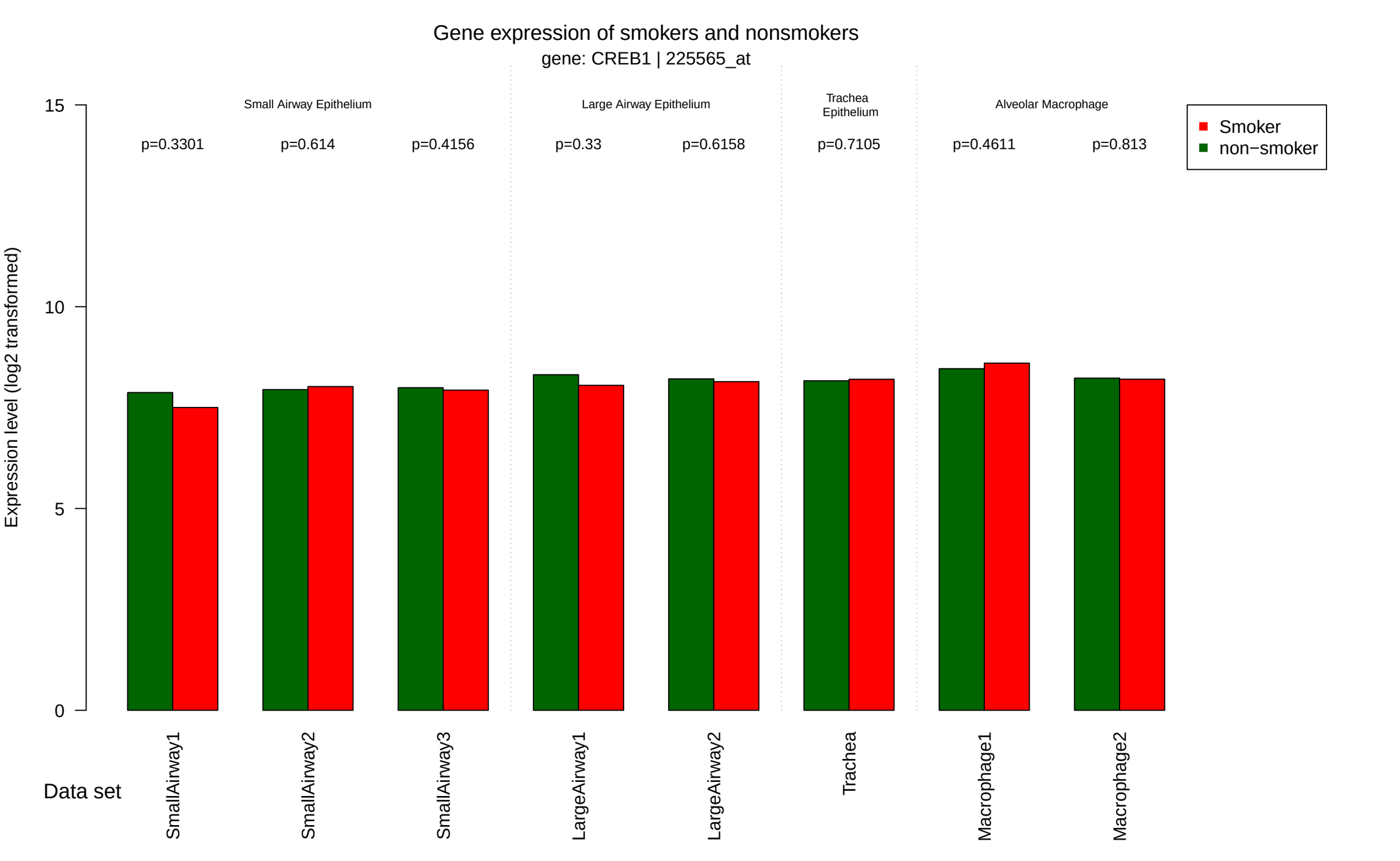
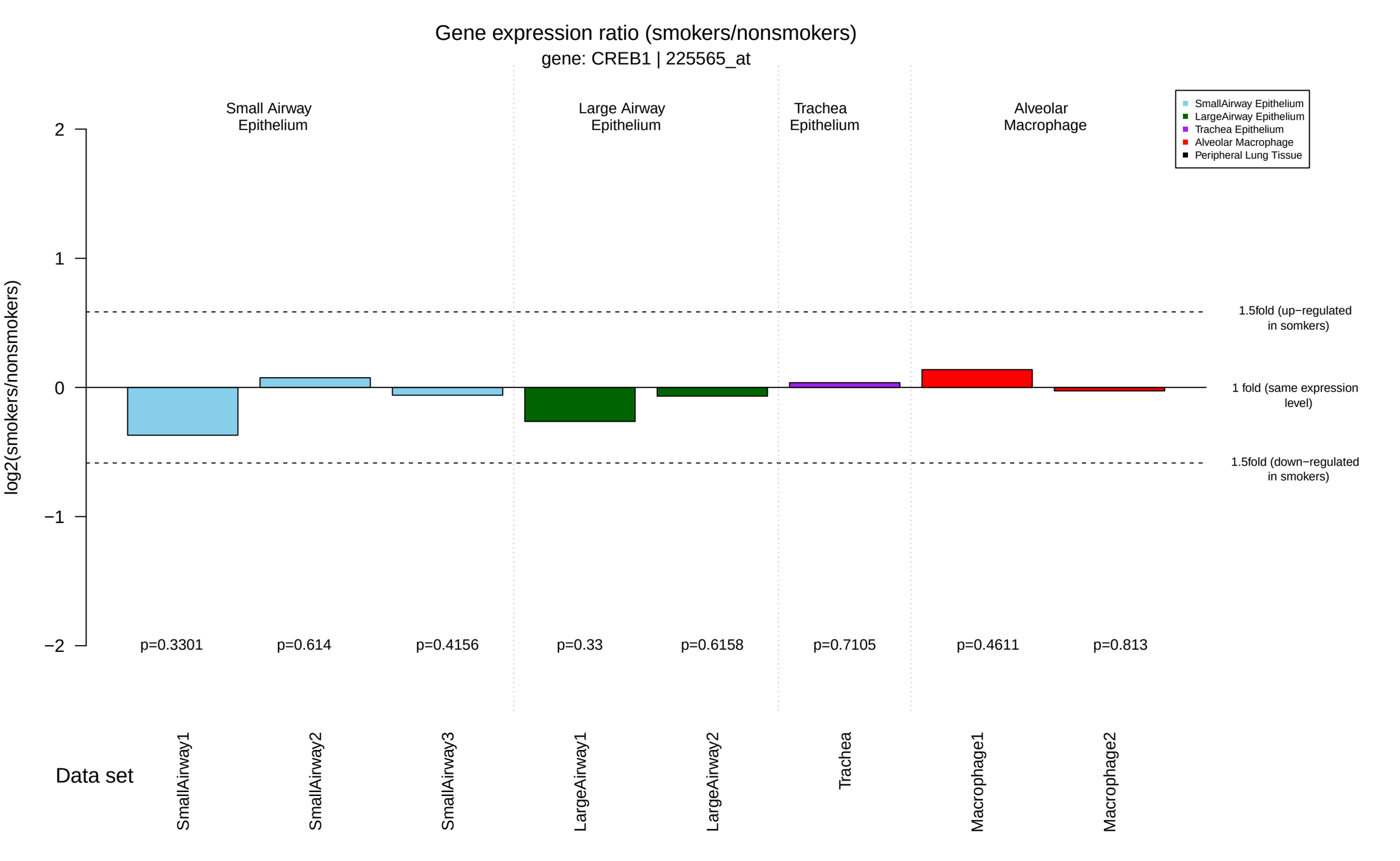
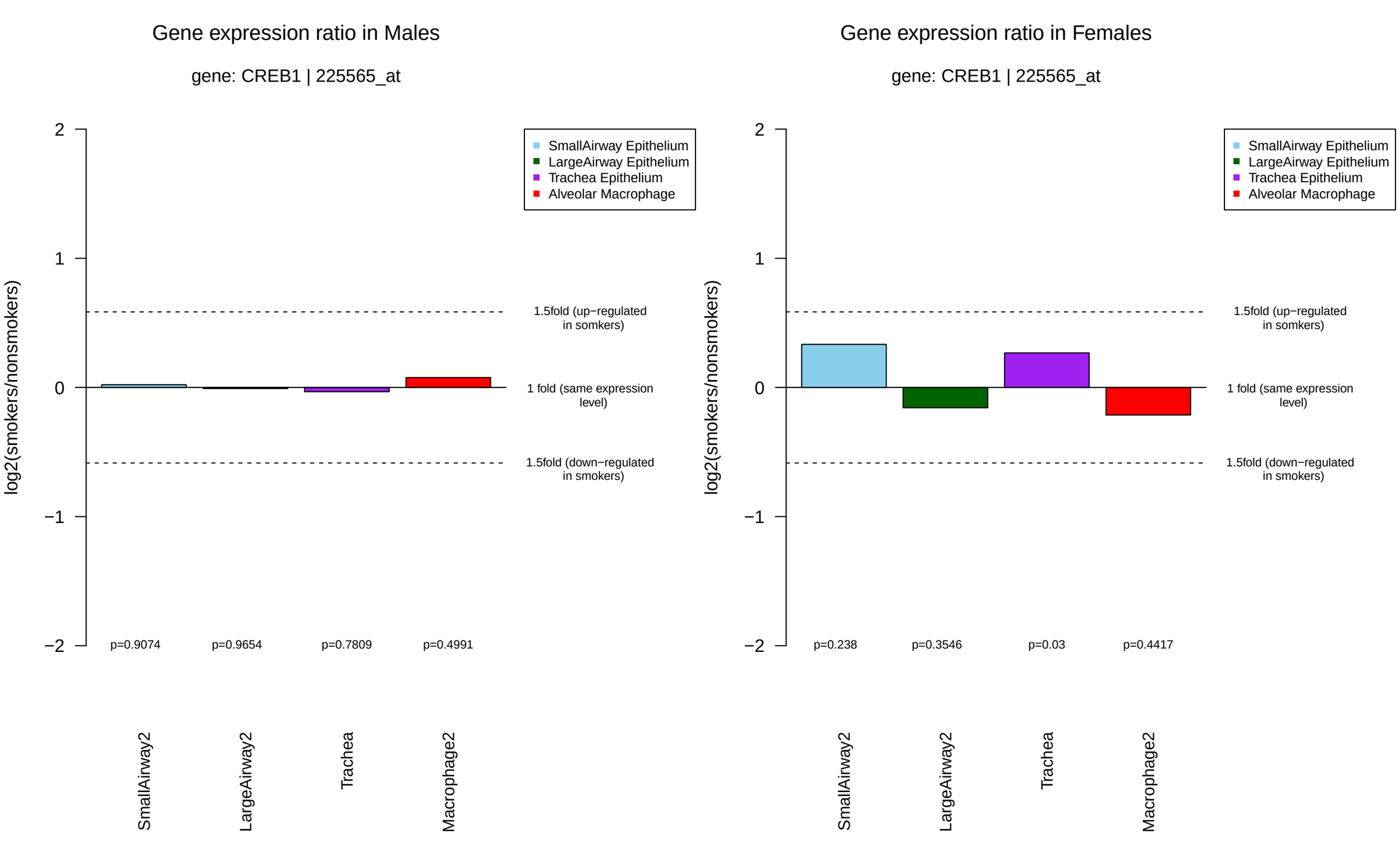
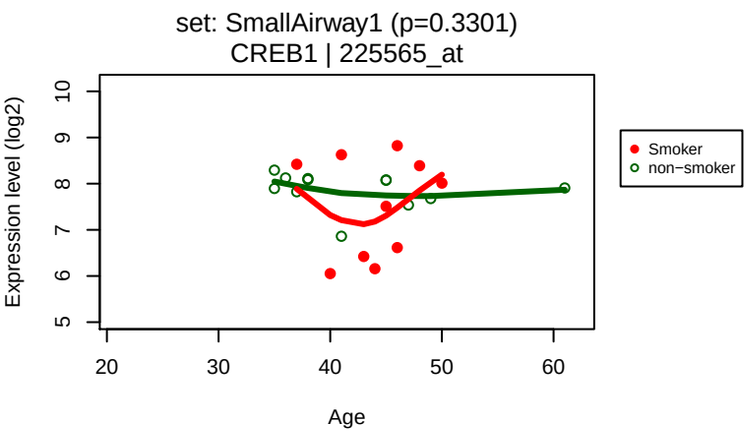
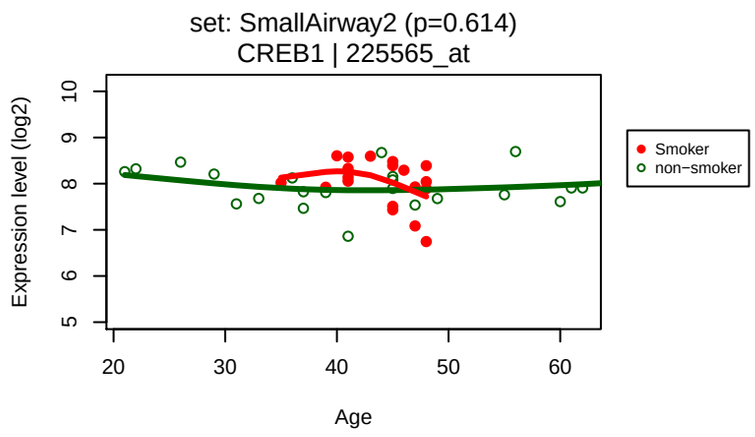
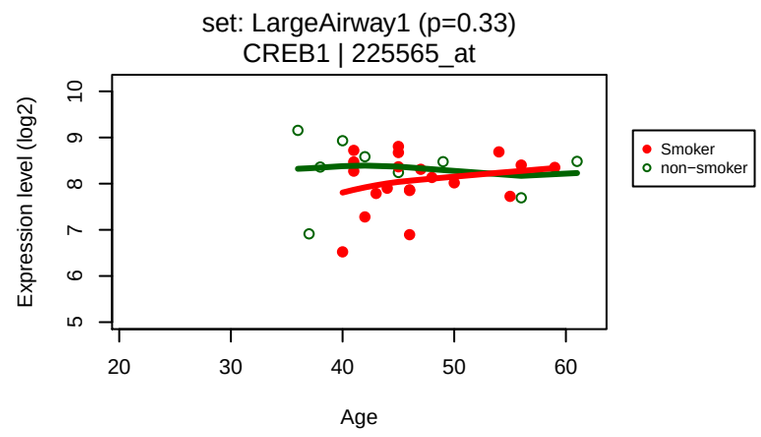

Gene expression of smokers and nonsmokers
gene: CREB1 | 225565_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3301
p=0.614
p=0.4156
p=0.33
p=0.6158
p=0.7105
p=0.4611
p=0.813
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

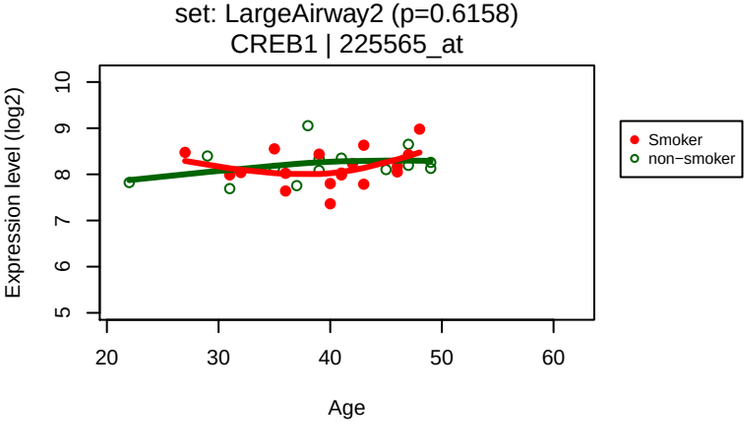

Gene expression ratio (smokers/nonsmokers)
gene: CREB1 | 225565_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3301
p=0.614
p=0.4156
p=0.33
p=0.6158
p=0.7105
p=0.4611
p=0.813
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

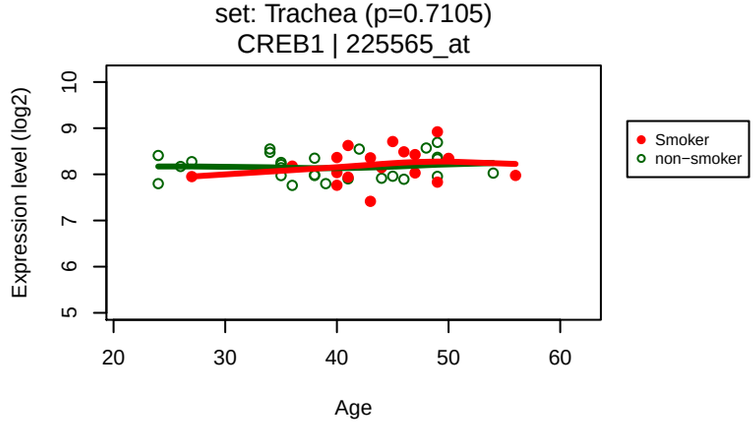

Gene expression ratio in Males
gene: CREB1 | 225565_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9074
p=0.9654
p=0.7809
p=0.4991
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CREB1 | 225565_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.238
p=0.3546
p=0.03
p=0.4417
SmallAirway2
LargeAirway2
Trachea
Macrophage2

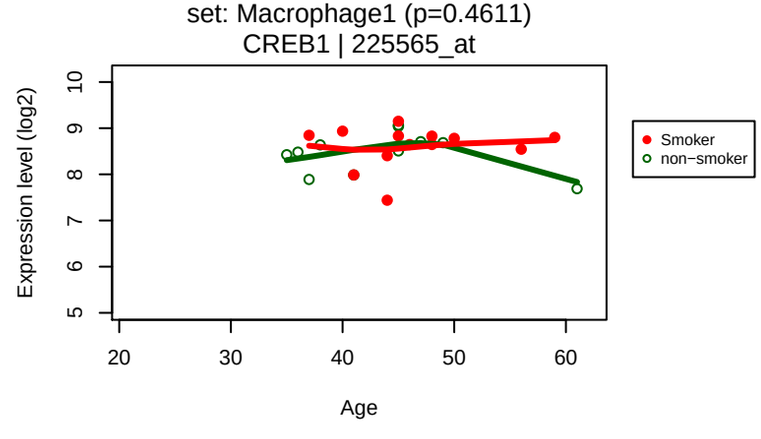

set: SmallAirway1 (p=0.3301)
CREB1 | 225565_at
Expression level (log2)
Age
Smoker
non−smoker

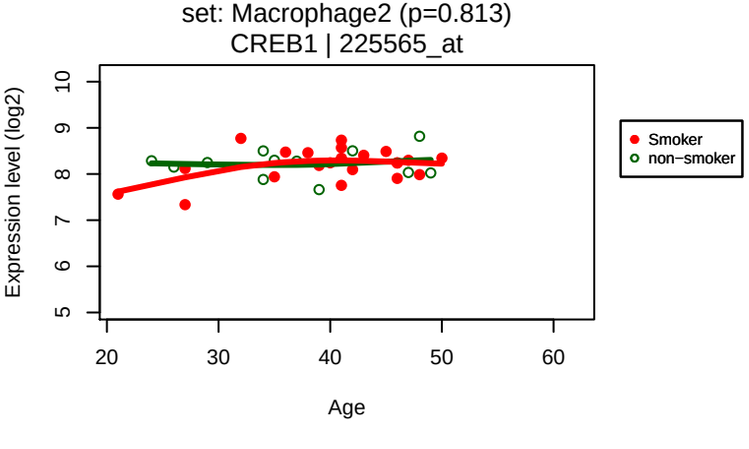

set: SmallAirway2 (p=0.614)
CREB1 | 225565_at
Expression level (log2)
Age
Smoker
non−smoker

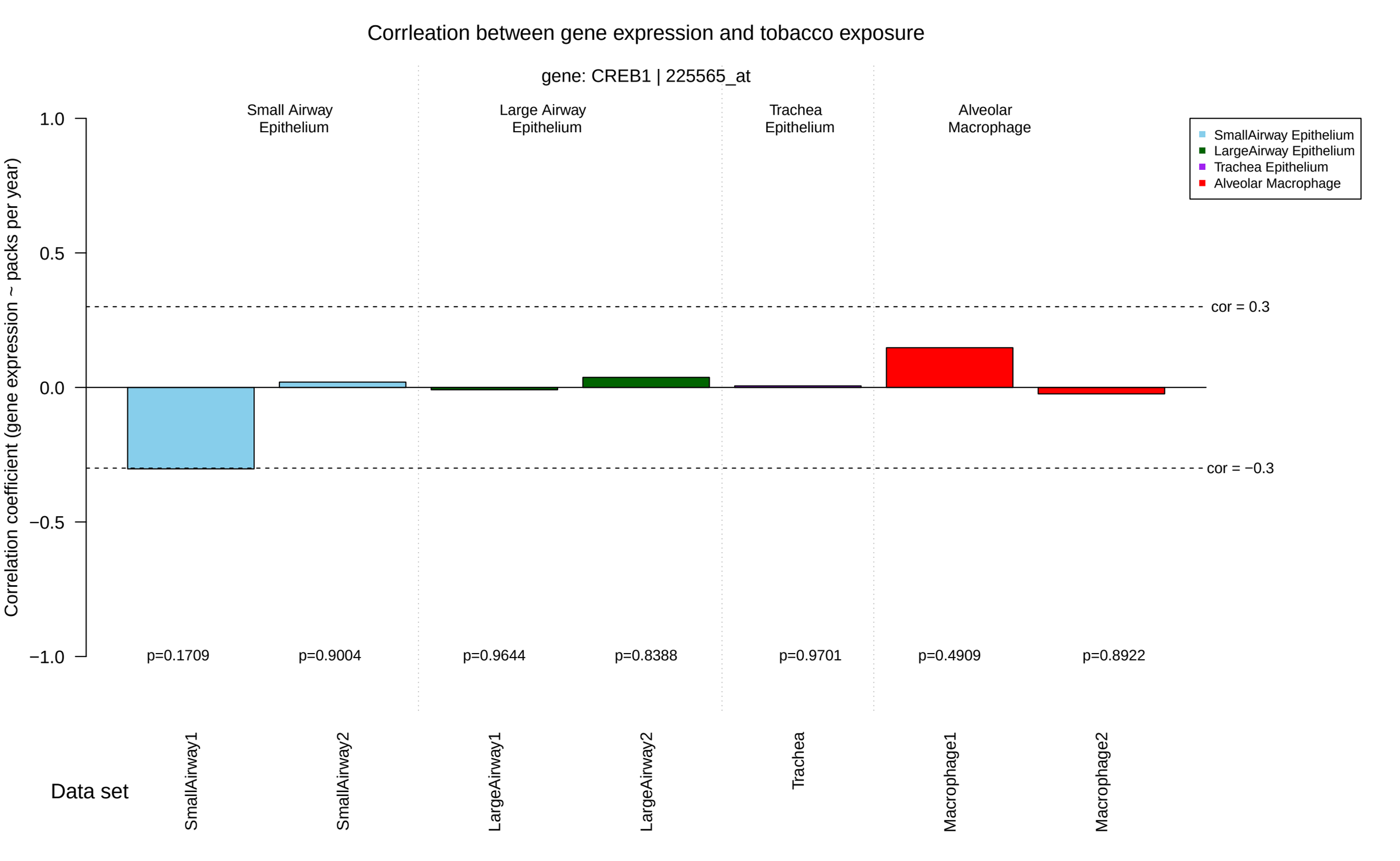

set: LargeAirway1 (p=0.33)
CREB1 | 225565_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6158)
CREB1 | 225565_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7105)
CREB1 | 225565_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4611)
CREB1 | 225565_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.813)
CREB1 | 225565_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CREB1 | 225565_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1709
p=0.9004
p=0.9644
p=0.8388
p=0.9701
p=0.4909
p=0.8922
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2